

Experiment No. 6

Experiment Name

Predictive Analytics using Linear Regression and Model Performance Evaluation

Aim

To develop a Linear Regression model using Python for predicting continuous outcomes and evaluate its performance using appropriate regression metrics.

Purpose

Predictive analytics uses historical data and machine learning algorithms to forecast future outcomes and support strategic decision-making. Linear Regression is one of the most widely used supervised learning algorithms for predicting continuous numerical values by establishing relationships between independent and dependent variables.

In this experiment, students will use datasets such as the House Price Prediction Dataset, Advertising Sales Dataset, or California Housing Dataset to build a regression model, make predictions, and evaluate its accuracy using statistical performance measures.

Prerequisites

- Basic knowledge of Python programming
- Familiarity with Pandas, NumPy, Matplotlib, and Scikit-learn
- Understanding of supervised machine learning and regression concepts
- Knowledge of data preprocessing and Exploratory Data Analysis (EDA)
- Jupyter Notebook or Google Colab

Steps

1. Select a suitable dataset such as the House Price Prediction Dataset, Advertising Sales Dataset, or California Housing Dataset.
2. Import the required libraries including Pandas, NumPy, Matplotlib, and Scikit-learn, and load the dataset.
3. Perform basic data preprocessing by handling missing values, encoding categorical variables (if any), and selecting relevant features.
4. Split the dataset into training and testing sets using an appropriate train-test ratio (e.g., 80:20).
5. Train a Linear Regression model using the training dataset.

6. Use the trained model to predict values for the testing dataset.
7. Visualize the regression line and compare the actual and predicted values using appropriate plots.
8. Evaluate the model using performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score.
9. Analyze the regression coefficients to understand the impact of each independent variable on the target variable.
10. Interpret the results and discuss the suitability of Linear Regression for the selected prediction problem.

Questions

1. What is Predictive Analytics, and how is it used in real-world applications?
2. Explain the working principle of the Linear Regression algorithm.
3. Differentiate between dependent and independent variables with suitable examples.
4. Why is it necessary to split the dataset into training and testing sets?
5. What is the significance of the R^2 Score in regression analysis?
6. Differentiate between MAE, MSE, and RMSE. Which metric is more sensitive to large prediction errors?
7. What assumptions should be satisfied before applying Linear Regression?
8. How can overfitting and underfitting affect the performance of a regression model?
9. Mention any three real-world applications of Linear Regression in business or industry.
10. How can feature selection improve the accuracy and interpretability of a predictive model?